

If you're not getting the same results, try changing the hyperparameters, particularly the learning rate, mini-batch size, maximum episode length, and minimum forward progress. Training for 5,000 epochs with intrinsic rewards works, but in our experience it's sensitive to these hyperparameters. Of course, 5,000 epochs is not very long, so training for longer will result in more interesting behavior.

How will this work in other environments?

We trained our DQN agent with an ICM-based reward on a single environment, Super Mario Bros, but the paper "Large-Scale Study of Curiosity-Driven Learning" by Yuri Burda et al. (2018) demonstrated how effective intrinsic rewards alone can be. They ran a number of experiments using curiosity-based rewards across multiple games, finding that a curious agent could progress through 11 levels in Super Mario Bros. and could learn to play Pong, among other games. They used essentially the same